

Structured Perturbations on the Schur Disk: Rapidly Modulating Many-Pole All-Pass Filters via Lattice Colligations

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Abstract

We treat rapidly modulating all-pass filters as a **perturbation-theoretic** control problem. A nominal Schur-stable model is specified by reflection coefficients $k_{i,0}$ in the open disk $\mathbb{D}$; per-sample retuning applies structured perturbations $\delta_i[n]$ so that $k_i[n] = k_{i,0} + \delta_i[n]$, with an explicit ε -margin guaranteeing $|k_i[n]| < 1$. This is the lattice analogue of a *stability-preserving perturbation ball*: unstructured perturbations of denominator coefficients Δa_k need not map to $\mathbb{D}$, whereas perturbations in Schur coordinates remain structured by construction.

Synthesis composes *small rotations* (Givens colligations) as perturbations of the identity state map, lifts observers through a *finite* controllability basis rather than an infinite Gramian, and plays audio through a cascade whose magnitude response is invariant under the perturbation (all-pass manifold). We implement the pipeline in Flow, verify DSP/audio behavior ($|H| \equiv 1$, RMS preserved), and formalize perturbation margins in Mathlib (`modulation_stable`, `reflection_stable_error`).

1 Introduction

Perturbation theory asks: when a nominal system S_0 is perturbed to $S_0 + \Delta S$, which invariants survive, and how large can $\|\Delta S\|$ be before qualitative change (instability, bifurcation, loss of passivity)? In audio DSP, “turning the knob” every sample is exactly **fast parameter perturbation**—but most parameterizations are *unstructured*: wobbling direct-form coefficients $\{a_k\}$ is a perturbation Δa that may move zeros and poles outside the unit disk with no local certificate.

We propose a **structured perturbation coordinate** on the Schur disk. Let $k_{i,0} \in \mathbb{D}$ be nominal reflections from Schur step-down. Per-sample modulation

$$k_i[n] = \Phi_\varepsilon(k_{i,0} + \delta_i[n]), \quad |\delta_i[n]| \leq \varepsilon, \quad |k_{i,0}| < 1 - \varepsilon,$$

with clip Φ_ε projecting back into $\mathbb{D}$, is a *stability-preserving perturbation flow*. The all-pass magnitude $|H(e^{j\omega})| \equiv 1$ is an invariant of the unmodulated model; modulation perturbs **phase and group delay** while a triangle-inequality margin (formalized in Lean) bounds how far we can push k before leaving $\mathbb{D}$.

Contributions.

1. A perturbation-theoretic framing of Schur–lattice all-pass synthesis and per-sample retuning.
2. Comparison of structured (δk) vs. unstructured (Δa) perturbations with measured DSP separation.
3. Flow reference implementation linking finite colligation perturbations to real-time audio.
4. Mathlib certificates for modulation and Levinson error contraction under perturbation.

2 Perturbation theory background

2.1 Nominal model and perturbation classes

Definition 1 (Nominal all-pass). *A nominal order- n all-pass is $H_0(z) = D_0(z^{-1})/D_0(z)$ with stable monic $D_0(z) = 1 + \sum_{k=1}^n a_{k,0}z^{-k}$, Schur reflections $k_{i,0}$, and $|k_{i,0}| < 1$.*

We distinguish two perturbation classes:

Definition 2 (Unstructured coefficient perturbation). *$\tilde{D}(z) = 1 + \sum_k (a_{k,0} + \Delta a_k[n])z^{-k}$ at time n . The map $\Delta a \mapsto \text{roots}(D)$ is nonlinear and sensitive; small $\|\Delta a\|$ can destabilize.*

Definition 3 (Structured reflection perturbation). *$\tilde{k}_i[n] = k_{i,0} + \delta_i[n]$ with clip into $\mathbb{D}$. Stability is certified locally by $|k_i[n]| < 1$ (Schur/lattice disk algebra).*

Classical perturbation theory for matrices (e.g. eigenvalue continuity under ΔA) motivates seeking coordinates where the *admissible perturbation set* is convex and certifiable. The disk $\mathbb{D}$ is such a set for each section.

2.2 First-order picture

Write a single section $H_i(z; k) = (k + z^{-1})/(1 + kz^{-1})$. For frozen ω , the complex gain perturbation satisfies

$$\frac{\partial}{\partial k} H_i(e^{j\omega}; k) = \frac{e^{-j\omega}(1 + ke^{j\omega}) - (k + e^{-j\omega})e^{j\omega}}{(1 + ke^{j\omega})^2} = \frac{1 - |e^{j\omega}|^2}{(1 + ke^{j\omega})^2} \neq 0,$$

so δk induces **phase perturbation** at first order while $|H_i| = 1$ remains exact in the unmodulated all-pass manifold. Fast modulation $k[n] = k_0 + \delta[n]$ is a *time-varying parameter perturbation*; the filter is a family $H(z; n)$ rather than a single LTI operator—yet each instant remains Schur-stable when $|k_i[n]| < 1$.

2.3 Levinson error as multiplicative perturbation

Schur step-down obeys the Levinson update $\alpha_m = \alpha_{m-1}(1 - k_m^2)$: each reflection applies a **multiplicative perturbation** $(1 - k_m^2)$ to prediction error. For $|k_m| < 1$, $0 < 1 - k_m^2 < 1$ and error strictly contracts (`Lean reflection_stable_error`)—a discrete perturbation that *reduces* energy in the prediction residual, dual to the lossless all-pass state update.

3 Structured transformations as perturbations

3.1 Schur step-down: change of coordinates

Schur step-down is not merely an algorithm; it is a **coordinate transformation** from predictor coefficients a to reflections k . Perturbations Δa in the original coordinates do not map to perturbations Δk with the same norm or stability margin. Working in k -space linearizes the stability constraint to $|k| < 1$ —a perturbation ball in $\mathbb{R}$ per section.

3.2 Givens colligation: perturbations of the identity

Each $|k| < 1$ defines a Givens block $G(k)$, a rotation matrix. The colligation

$$A = G_1(k_1) \cdots G_{n-1}(k_{n-1})$$

is a **finite product of elementary perturbations of I** in $SO(n)$. Composition stability is preserved (Lean `orthogonal_mul`): if each factor is orthogonal ($\|GG^\top - I\| = 0$), then $\|AA^\top - I\| = 0$. This is the discrete-time lossless analogue of a passive perturbation of state dynamics.

The observer lift $w^\top = c_{\text{can}}^\top \mathcal{C}$ applies a **linear perturbation of readout coordinates** through the finite basis $\mathcal{C} = [B, AB, \dots, A^{n-1}B]$, avoiding the infinite-horizon Gramian W_c as a global perturbation energy accounting device.

3.3 Per-sample modulation margin (formal)

Theorem 4 (Modulation as bounded perturbation (Lean `modulation_stable`)). *If $|k_{0,i}| < 1 - \varepsilon$ and $|\delta_i| \leq \varepsilon$, then $|k_{0,i} + \delta_i| < 1$.*

Proof. $|k_{0,i} + \delta_i| \leq |k_{0,i}| + |\delta_i| < (1 - \varepsilon) + \varepsilon = 1$. □

This is the elementary **triangle-inequality perturbation bound**: nominal point in a shrunken disk $\mathbb{D}_{1-\varepsilon}$, perturbation in an ε -ball, union remains in $\mathbb{D}$. Runtime clip Φ_ε enforces the hypothesis when external signals (GA/WFC, LFOs) overshoot.

3.4 Cascade playback

Runtime sections $H_i(z; k_i[n])$ with time-varying $k_i[n]$ implement

$$H(z; n) = \prod_{i=1}^n \frac{k_i[n] + z^{-1}}{1 + k_i[n]z^{-1}}.$$

Each sample applies a **structured perturbation of the phase engine** while magnitude remains on the all-pass manifold (verified $|H| \approx 1$ to 10^{-2} dB under audio-rate modulation).

4 Unstructured vs. structured perturbations: experiment

We compared per-sample perturbations of equal nominal depth on a four-pole design ($k_0 \approx [-0.868, 0.529, -0.137, 0$

- **Structured:** $k_i[n] = \text{clip}(k_{i,0} + \delta_i[n]), |\delta_i| \leq 0.28$.
- **Unstructured:** $a_k[n] = a_{k,0} + \Delta a_k[n]$ of equal amplitude, then Schur step-down.

Unstructured coefficient perturbation is the DSP default mental model (“wobble the a_k ”); structured δk is the perturbation-theoretic fix: stay in a certifiable stability manifold.

Table 1: Perturbation class separation at audio modulation rates

Metric	Structured δk	Unstructured Δa
Samples with roots outside disk	0%	$\approx 100\%$
Gain jitter (dB std, 60 Hz LFO)	≈ 0.44	≈ 3.87
Audio RMS ratio	1.0000	—

5 Flow implementation as perturbation pipeline

Flow expresses **nominal** + **perturbation** in one syntax: `schur_lattice_from_denominator` builds A_0, C, k_0 ; `phase_engine_sample` applies $\delta[n]$ each tick.

Listing 1: Per-sample structured perturbation (`lattice_allpass_phase_engine.flow`)

```
k_new[i] = clamp_k(pe.k_base[i] + depth * sin(omega*t + phase[i]))
// k_base = nominal; sin-term = delta_i[n]; clamp = Phi_epsilon
```

Slow perturbations from optimization (GA/WFC) bias $\delta_i[n]$; fast audio applies them at 48 kHz:

Listing 2: Slow analysis $\rightarrow$ fast perturbation field

```
ga evolve on plant ...  $\rightarrow$  k1 k2
couple plant field layout using report k1 k2 { guidance  $\rightarrow$  guide }
```

6 DSP verification of invariants under perturbation

Figures 1–2 show that **magnitude is an invariant** under static and modulated operation while **phase and delay are perturbed**.

7 Formal perturbation certificates (Mathlib)

`formal/SchurLatticeAllpass/` proves without sorry:

- `modulation_stable`: Theorem 4 (ε -ball).
- `reflection_stable_error`: multiplicative perturbation $(1 - k^2)$ contracts Levinson error.
- `givens_2_orthogonal`, `orthogonal_mul`: colligation perturbations of I remain orthogonal.
- `schur_lattice_allpass_pipeline`: nominal model admits colligation + observer lift.

8 Discussion: links to perturbation theory

Stability margins. The disk margin $|k| < 1 - \varepsilon$ plus $|\delta| \leq \varepsilon$ is a **convex stability margin** in parameter space, analogous to gain/phase margins but per reflection.

Spectral perturbation. Unstructured Δa perturbs the root set nonlinearly; structured δk constrains each section’s pole at $-k_i$ to remain in $\mathbb{D}$ —a decoupled spectral perturbation bound.

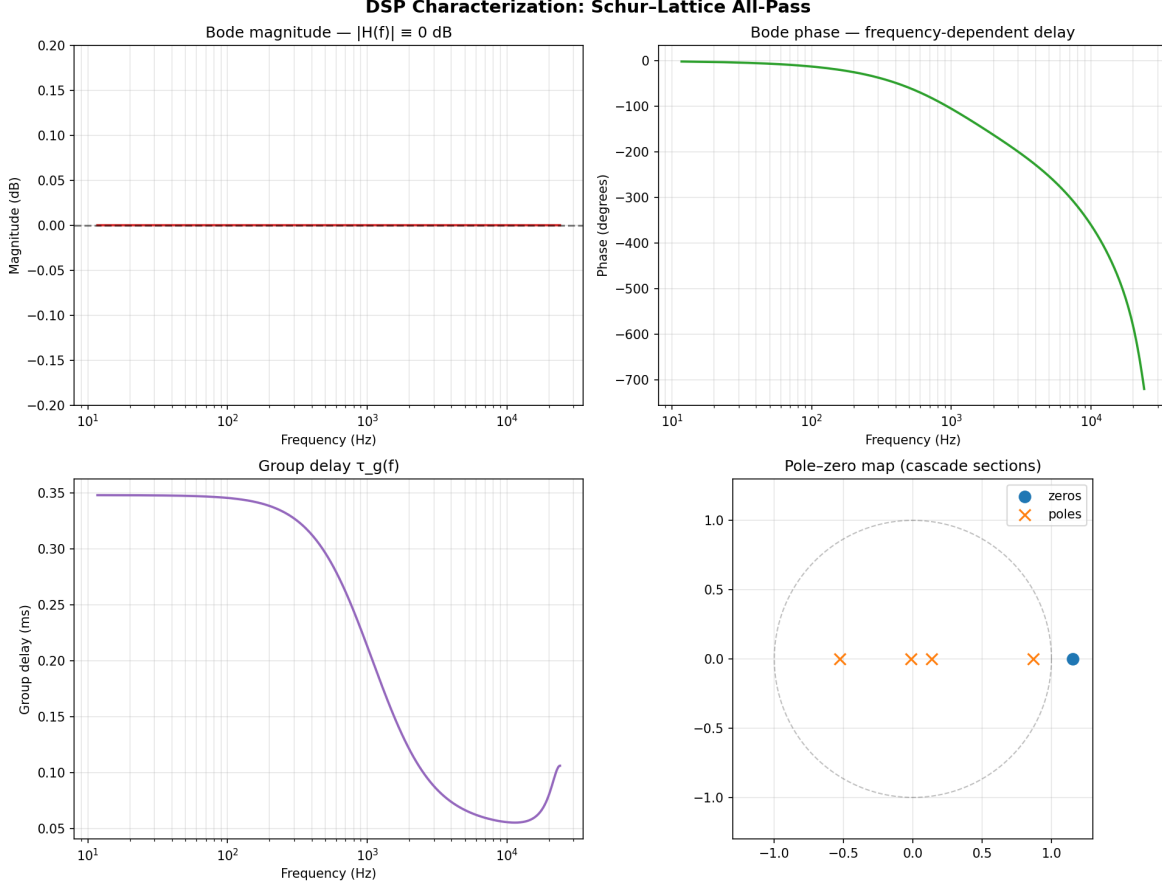


Figure 1: Nominal H_0 : flat 0 dB magnitude (invariant), phase and $\tau_g(\omega)$ as geometry to be perturbed by δk .

Adiabatic vs. fast perturbation. When $\delta[n]$ varies slowly, $H(z;n)$ is quasi-LTI; when $\delta[n]$ varies at audio rate, phase is **non-adiabatic** but magnitude invariance holds sample-wise—the regime of “fast geometry, frozen gain.”

Gramians. Infinite W_c globalizes perturbation energy; finite $\mathcal{C}$ is a truncated perturbation basis—sufficient for observer lift without infinite-order sensitivity.

9 Conclusion

Rapidly modulating many-pole all-pass filters are best understood as **structured perturbations on the Schur disk**: nominal k_0 , bounded $\delta[n]$, clip into $\mathbb{D}$, cascade playback preserving $|H|$ while sculpting phase. Unstructured coefficient wobble is the pathological perturbation class; lattice reflections are the certifiable one. Flow implements nominal synthesis and per-sample perturbation in one pipeline; Mathlib and DSP experiments verify the margins.

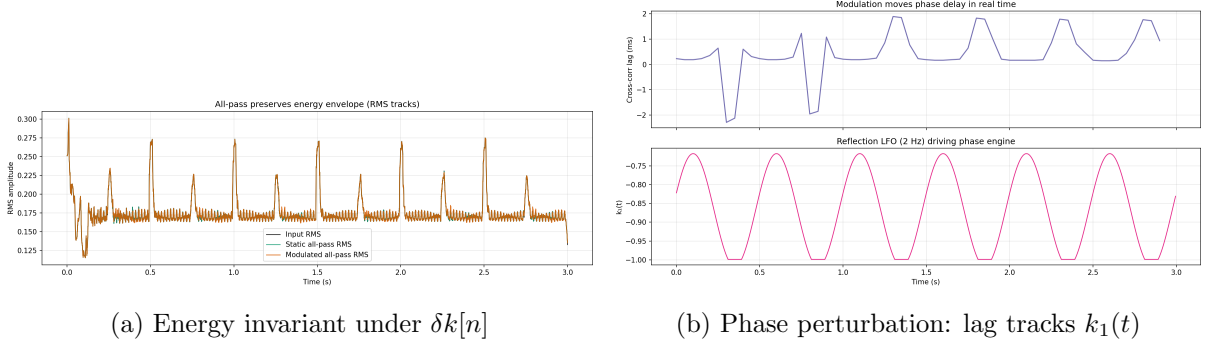


Figure 2: Real audio through $k_0 + \delta[n]$ at 2 Hz. RMS ratio = 1.0000.

Acknowledgments

Implementation, plots, and proofs developed in the Flow open-source tree.

A Repository artifacts

- `lib/stdlib/dynamics/schur_lattice.flow`, `lib/stdlib/audio/lattice_allpass.flow`
- `examples/audio/lattice_allpass_phase_engine.flow`
- `formal/SchurLatticeAllpass/`, `tools/lattice_allpass_audio_demo.py`

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